The union of a set of cardinals is a cardinal

The Open Logic Project contributors

openlogic:set-theory/cardinals:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own proposition environment unioncardinalscardinal (source lines 691-693, the \ollabel on line 691); the proof environment begins two lines below it and is its own proof, closing at source line 703.

Editorial note. Neither the statement nor the proof contains an \olref or a \cite, so no source pointer and no bibliography entry are needed; the cardinal-comparison notation used in the proof is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained proposition prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's set-theory/cardinals unit, counted once: the union of a set of cardinals is a cardinal.

Allowed standard prerequisites (editorial).

- Cardinal. A cardinal is an ordinal that is not equinumerous with any of its elements.. The proof puts alpha in a member b of X that is a cardinal and uses that fact to get alpha is smaller than b (source lines 697-699), and the conclusion is exactly that union X is not equinumerous with any of its elements (source lines 701-702).
- Cardinal comparison. For ordinals alpha and b, alpha is less than b as a cardinal when alpha is equinumerous with a member of b; the relation is transitive and behaves monotonically under subsets.. The chain alpha < b <= union X, hence alpha != union X, is the core of the proof (source lines 698-701).

Proposition 0.1. *If every member of X is a cardinal, then $\bigcup X$ is a cardinal.*

Proof. It is easy to check that $\bigcup X$ is an ordinal. Let $\alpha \in \bigcup X$ be an ordinal; then $\alpha \in \mathfrak{c}b \in X$ for some cardinal $\mathfrak{c}b$. Since $\mathfrak{c}b$ is a cardinal, $< \alpha\mathfrak{c}b$. Since $\mathfrak{c}b \subseteq \bigcup X$, we have $\leq \mathfrak{c}b\bigcup X$, and so $\neq \alpha\bigcup X$. Generalising, $\bigcup X$ is a cardinal. □